Problem Set 2
Due Date: Wednesday November 18th

Please submit your answers at the start of class. Physical copies are preferred, but digital submissions are acceptable as well. Please show your work, that is to say, be clear about which derivatives you are taking, how terms cancel in the algebra, etc.

Problem 1: Finishing up Misallocation
Take the misallocation framework from class. Prove the following:

1. Define $L = \sum_i L_i$, $K = \sum_i K_i$, and $Y = \left(\sum_i y_i^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}$. Define aggregate TFP as

$$Y = TFP \times K^{\alpha} L^{1-\alpha}.$$

   Prove that

$$TFP = \left[\sum_i \left(A_i \times \frac{\overline{TFPR}}{TFPR_i}\right)^{\sigma-1}\right]^{\frac{1}{\sigma-1}},$$

   where $\overline{TFPR}$ is the average TFPR.

2. Suppose that $TFPQ_i$ and $TFPR_i$ are jointly log-normal. Prove that aggregate TFP satisfies

$$\log TFP = \frac{1}{\sigma-1}\log\left(\sum_i (TFPQ_i)^{\sigma-1}\right) - \frac{\sigma}{2}\text{Var}(\log TFPR_i)$$

Problem 2: Alternative Sources for Variable Markups
In this problem, we explore an alternative source of variable markups: differing demand elasticities. To do so, we introduce the Kimball aggregator. Consider a continuum of differentiated varieties $i \in [0, 1]$. A representative household chooses quantities q_i of each good, where aggregate consumption is defined implicitly as

$$\int_0^1 G\left(\frac{q_i}{Q}\right) di = 1$$

where $G : \mathbb{R} \to \mathbb{R}$ is an increasing, strictly concave function with $G(1) = G'(1) = 1$.

Households take prices p_i as given, and chooses consumption to minimize expenditure subject to aggregate consumption is equal to Q.

1. Write down the household's expenditure minimization problem, and derive the FOC for good i.

2. Derive the aggregate price index such that the FOC can be written

$$\frac{p_i}{P} = G'\left(\frac{q_i}{Q}\right).$$

   Show that it satisfies

$$P = \int_0^1 p_i \frac{q_i}{Q} di.$$

To make our lives slightly easier, let us consider the following functional form:

$$G(x) = 1 + \frac{1 - e^{-\alpha(x-1)}}{\alpha}, \qquad 0 < \alpha < 1.$$

3. Derive the demand for good i as a function of Q, p_i, and P.

4. Check that when prices are identical, $p_i = P$, that the symmetric allocation $q_i = Q$ solves the consumer's problem.

5. Define the price elasticity of demand for good (i) as $\varepsilon_i = -\frac{\partial \ln q_i}{\partial \ln p_i}$. Derive an expression for ε_i as a function of q_i/Q. Compare this to CES.

6. Suppose now that each product is produced by a monopolistically competitive firm with constant marginal cost φ_i, where wages are normalized to 1. Firms take P and Q as given when choosing their prices and quantities. Show that the firm's markup satisfies $\frac{\varepsilon_i}{\varepsilon_i - 1}$, and use the elasticity derived earlier to express it as a function of q_i/Q.

7. Explain drivers of variation in firm-level markups under Kimball preferences.

Problem 3: Two Country Atkeson Burstein

In this problem, we consider a two-country alternative to Atkeson and Burstein (2008). Suppose there are two countries Home and Foreign, and in one particular sector there are 4 home firms and 4 foreign firms. We study this sector, so assume that (exogenous) demand shifter in home and foreign is $E^H = E^F = 100$ such that

$$q_{ij}^H = p_{ij}^{-\sigma} P_j^{\sigma-\rho} E^H.$$

There are iceberg trade costs of $\tau = 1.2$. The vector of productivities describing home firms is $\{\varphi_i^H\}_{i=1,2,3,4} = \{1.5, 1.35, 1.25, 1.2\}$ and for foreign firms is $\{\varphi_i^F\}_{i=1,2,3,4} = \{1.55, 1.5, 1.45, 1.25\}$.

Suppose that the elasticity of substitution between sectors is $\rho = 4$, and within a sector is $\sigma = 10$. For now, assume that fixed costs of domestic production and fixed costs of exporting are equal to zero. "Normalize" the wage to be equal to 1 in each country.

1. Compute market shares $\{s_i^H, s_i^F\}_{i=1,2,3,4}$, markups $\{\mu_i^H, \mu_i^F\}_{i=1,2,3,4}$, revenues, profits, and the price indices in home and foreign for the fixed $\tau = 1.2$.

2. Compute a measure of market concentration, the Herfindahl-Hirschman Index (HHI), defined as:

$$HHI^H = \sum_{i=1,2,3,4, x=H,F} (s_i^x)^2$$

   where s_i^X is the market share of firm i from country X. Note that we include market shares in the domestic market by foreign firms in this computation. The HHI is a number between 0 and 1, where 1 means the market is dominated by 1 firm and 0 means the market is approximately perfect competition. An HHI of $1/N$ means the market is about as concentrated as a market with N equally sized firms.

3. Recompute market shares and the HHI in the domestic market as you vary τ from 1 to 2, and plot the result.

4. Does trade liberalization (that is to say, a reduction in τ) increase or decrease competition?

Suppose now that firms in both countries face entry costs $f_d = 2.4$ to produce in the local market, and $f_x = 2.55$ to export to the other market.

5. Show that the equilibrium where all 4 home firms operate domestically and 2 of the 4 foreign firms export to home is a valid equilibrium. That is to say, show that all firms when pricing optimally have positive profits and optimally select into operating in the relevant markets. How many home firms export in this case?

6. Now show that an equilibrium where 3 home firms operate domestically, and 3 of the 4 foreign firms export to home is also a valid equilibrium.

7. I will not make you check if there are other valid equilibria, but speculate on conditions in this general framework under which equilibrium multiplicity is a possibility.